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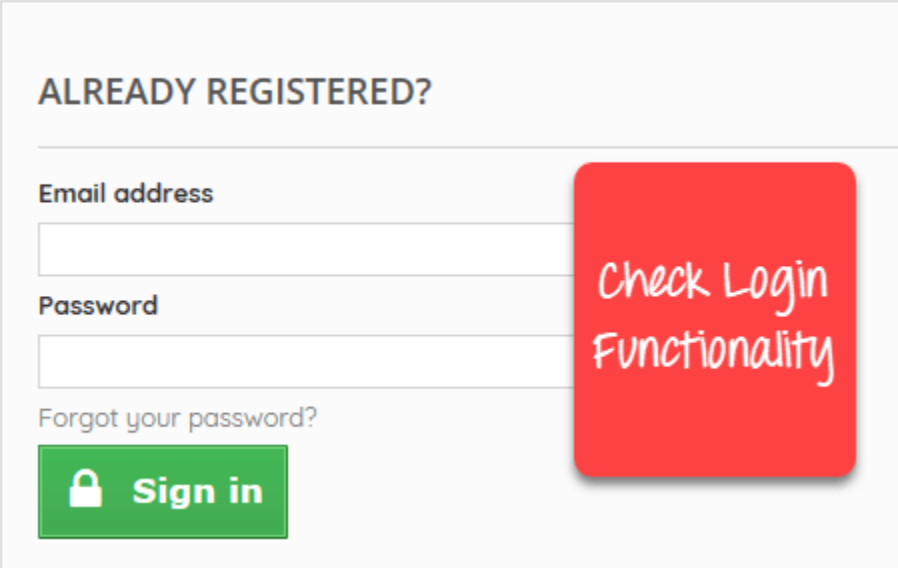
How to Write Test Cases with Examples

9–11 minutes

Steps to Create Test Cases in Manual Testing

Let's create a Test Case for the scenario: Check Login Functionality



Step 1) A simple test case to explain the scenario would be

Test Case #	Test Case Description
1	Check response when valid email and password is entered

Step 2) Test the Data.

In order to execute the test case, you would need Test Data.

Adding it below

Test Case #	Test Case Description	Test Data
1	Check response when valid email and password is entered	Email: guru99@email.com Password: lNf9^Oti7^2h

Identifying test data can be time-consuming and may sometimes require creating test data afresh. The reason it needs to be documented.

🥇 **RemotePC**

- ✔ Remote printing and session recording.
- ✔ Business-class online meetings with enhanced security.

🚀 **JIRA Software**

- ✔ Plan, track, and manage tasks efficiently.
- ✔ Enterprise-grade security for data protection.

👉 **Zoho BugTracker**

- ✔ Scalable and easy-to-use bug tracking system.

✔ Automation for workflows and notifications.

Step 3) Perform actions.

In order to execute a test case, a tester needs to perform a specific set of actions on the AUT. This is documented as below:

Test Case #	Test Case Description	Test Steps	Test Data
1	Check response when valid email and password is entered	1) Enter Email Address 2) Enter Password 3) Click Sign in	Email: guru99@email.com Password: lNf9^Oti7^2h

Many times the Test Steps are not simple as above, hence they need documentation. Also, the author of the test case may leave the organization or go on a vacation or is sick and off duty or is very busy with other critical tasks. A recently hire may be asked to execute the test case. Documented steps will help him and also facilitate reviews by other stakeholders.

Step 4) Check behavior of the AUT.

The goal of test cases in software testing is to check behavior of the AUT for an expected result. This needs to be documented as below

Test Case	Test Case Description	Test Data	Expected Result

#			
1	Check response when valid email and password is entered	Email: guru99@email.com Password: lNf9^Oti7^2h	Login should be successful

During test execution time, the tester will check expected results against actual results and assign a pass or fail status

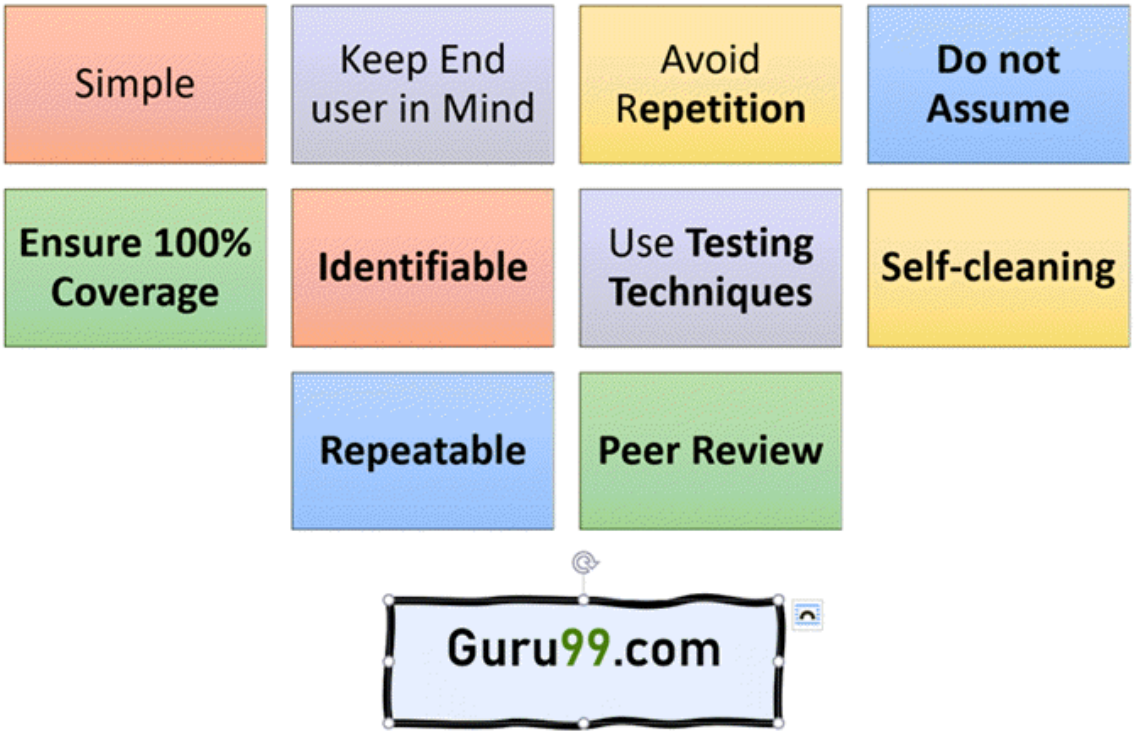
Test Case #	Test Case Description	Test Data	Expected Result	Actual Result
1	Check response when valid email and password is entered	Email: guru99@email.com Password: lNf9^Oti7^2h	Login should be successful	Login was successful

Step 5) That apart your test case -may have a field like,
Pre – Condition which specifies things that must be in place before the test can run. For our test case, a pre-condition would be to have a browser installed to have access to the site under test. A test case may also include Post – Conditions which specifies anything that applies after the test case completes. For our test case, a postcondition would be time & date of login is stored in the database

How to Write Test Case Video

Click here if the video is not accessible

Best Practice for writing good Test Case.



Test Case Best Practice

1. Test Cases need to be simple and transparent:

Create test cases that are as simple as possible. They must be clear and concise as the author of the test case may not execute them.

Use assertive language like go to the home page, enter data, click on this and so on. This makes the understanding the test steps easy and tests execution faster.

2. Create Test Case with End User in Mind

The ultimate goal of any software project is to create test cases that meet customer requirements and is easy to use and operate. A tester must create test cases keeping in mind the end user perspective

3. Avoid test case repetition.

Do not repeat test cases. If a test case is needed for executing some other test case, call the test case by its test case id in the pre-condition column

4. Do not Assume

Do not assume functionality and features of your software application while preparing test case. Stick to the Specification Documents.

5. Ensure 100% Coverage

Make sure you write test cases to check all software requirements mentioned in the specification document. Use Traceability Matrix to ensure no functions/conditions is left untested.

6. Test Cases must be identifiable.

Name the test case id such that they are identified easily while tracking defects or identifying a software requirement at a later stage.

7. Implement Testing Techniques

It's not possible to check every possible condition in your software application. Software Testing techniques help you select a few test cases with the maximum possibility of finding a defect.

- **Boundary Value Analysis (BVA):** As the name suggests it's the technique that defines the testing of boundaries for a specified range of values.
- **Equivalence Partition (EP):** This technique partitions the range into equal parts/groups that tend to have the same behavior.
- State Transition Technique: This method is used when software behavior changes from one state to another following particular

action.

- **Error Guessing Technique:** This is guessing/anticipating the error that may arise while doing manual testing. This is not a formal method and takes advantages of a tester's experience with the application

8. Self-cleaning

The test case you create must return the Test Environment to the pre-test state and should not render the test environment unusable. This is especially true for configuration testing.

9. Repeatable and self-standing

The test case should generate the same results every time no matter who tests it

10. Peer Review.

After creating test cases, get them reviewed by your colleagues. Your peers can uncover defects in your test case design, which you may easily miss.

While drafting a test case to include the following information

- The description of what requirement is being tested
- The explanation of how the system will be tested
- The test setup like a version of an application under test, software, data files, operating system, hardware, security access, physical or logical date, time of day, prerequisites such as other tests and any other setup information pertinent to the requirements being tested
- Inputs and outputs or actions and expected results

- Any proofs or attachments
- Use active case language
- Test Case should not be more than 15 steps
- An automated test script is commented with inputs, purpose and expected results
- The setup offers an alternative to pre-requisite tests
- With other tests, it should be an incorrect business scenario order

Test Case Management Tools

Test management tools are the automation tools that help to manage and maintain the Test Cases. Main Features of a test case management tool are

1. **For documenting Test Cases:** With tools, you can expedite Test Case creation with use of templates
2. **Execute the Test Case and Record the results:** Test Case can be executed through the tools and results obtained can be easily recorded.
3. **Automate the Defect Tracking:** Failed tests are automatically linked to the bug tracker, which in turn can be assigned to the developers and can be tracked by email notifications.
4. **Traceability:** Requirements, Test cases, Execution of Test cases are all interlinked through the tools, and each case can be traced to each other to check test coverage.
5. **Protecting Test Cases:** Test cases should be reusable and should be protected from being lost or corrupted due to poor version control. Test Case Management Tools offer features like

- Naming and numbering conventions
- Versioning
- Read-only storage
- Controlled access
- Off-site backup

Popular Test Management tools are: Quality Center and JIRA

The format of Standard Test Cases

Below is a format of a standard login Test cases example.

Test Case ID	Test Case Description	Test Steps	Test Data	Expected Results
TU01	Check Customer Login with valid Data	1. Go to site `http://demo.guru99.com` 2. Enter UserId 3. Enter Password 4. Click Submit	Userid = guru99 Password = pass99	User should Login into an application

Test Case ID	Test Case Description	Test Steps	Test Data	Expected Results
TU02	Check Customer Login with invalid Data	1. Go to site `http://demo.guru99.com` 2. Enter UserId 3. Enter Password 4. Click Submit	Userid = guru99 Password = glass99	User should not Login into an application

This entire table may be created in Word, Excel or any other Test management tool. That's all to Test Case Design

Test Case Template

- Please note that the template used will vary from project to project. Read this tutorial to Learn Test Case Template with Explanation of Important Fields

  Download the above Test Case Template Excel (.xls)

FAQs